ProcureSphere 360 - Frontend Architecture

Django Templates + HTMX with Bootstrap 5/Tailwind CSS
(Django Templates and React Components)

Client / User

- Super Admin
- Requester
- Department Approver
- Procurement Executive
- Procurement Manager
- Finance / AP
- Stores / Receiver
- Legal / Contract Manager
- Compliance / Auditor
- Vendor User

Access via Web Browser

Desktop
Tablet
Mobile

Frontend (User Interface Layer)

Application Shell (Django Templates)

ProcureSphere 360

- Dashboard
- Vendor Management
- Purchase Requisition
- Sourcing (RFQ/RFP)
- Purchase Order
- Receiving & Inspection
- Invoice & 3-Way Match
- Contract Management
- Budget & Spend
- Reports & Audit

Search... | User Profile

Main Content Area (Django Templates + HTMX)

Django Templates

- Page structure
- Layout and navigation
- Server-rendered HTML
- Integration with HTMX
- Role-based UI rendering

HTMX

- Partial page updates
- Dynamic tables
- Form submissions
- Modal loading
- Asynchronous interactions
- Real-time UI updates

Django Templates and React Components (for isolated, justified widgets)

- Interactive dashboards/charts (if required)
- Advanced data visualizations
- Complex UI components
- Implemented only where needed

Authentication & Session Management

- Login (Username/Password)
- Session Based Authentication (Django Sessions)

OR

- JWT Based Authentication (for API access if required)

Session / JWT via HTTPS

Backend Layer (Django + DRF)

Django Views (HTML responses) & REST APIs (JSON responses)
No changes to backend design

Frontend Assets & Structure

Templates

- Base templates
- Page templates
- Partial templates (for HTMX)

Static Assets

- CSS (Bootstrap 5 / Tailwind CSS)
- JavaScript (HTMX)
- Images, icons, fonts
- Custom JS (where required)

Reusable Components

- UI components (templates)
- HTMX partials
- Form components
- Table components
- Modal components
- Notification components

Django Templates and React Components

- Charts
- Advanced widgets
- Used only where needed

API Communication

- Django views via HTMX (HTML)
- REST APIs via AJAX/fetch (JSON)
- CSRF protection for form submissions
- Session or JWT based authentication (as per backend design)
- Secure communication over HTTPS